"RISE TO KNOWLEDGE"

¬trust the process see the Sunshine¬

OUR CADRE HAS SLOGGED A PARAGON

on

"Psycholinguistics (ENG511)"

for

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Special thanks to our mentor "Maha Malik"

COMPOSED BY



Lesson-09

- Sentence memory concerns retain the exact or verbatim wording of a sentence or simply its meaning.
- Fillenbaum presented people with a long list of **unrelated sentences**
- We ordinarily use the syntactic structure of a **sentence** to extract the underlying meaning.
- A classic study by Sachs examined the time parameters within which these processes might operate.
- The tendency to store only the meaning of a sentence in permanent memory is not limited to spoken languages.
- A few studies have examined the way pragmatic factors interact with semantic and syntactic considerations in sentence memory.
- The notion that greater elaboration of processing leads to better retention has received a substantial amount of support in psychological studies of words, sentences, and discourse.
- Elaboration is thought of as a process by which incoming information is related to information already stored in permanent memory.
- Information processing pertaining to the pragmatic functions of everyday speech may serve as the basis for elaboration.
- Elaborations based on our general knowledge of the world, is not specifically linguistic in nature.
- A particular form of elaborative processing is the drawing of inferences.
- The general experimental procedure used by Bransford has been to present people with long lists of sentences and later to probe their tendency to make false recognition errors:
- Johnson, Bransford, and Solomon examined people's comprehension and retention.
- The Deese–Roediger–McDermott (DRM) paradigm is a procedure in cognitive psychology used to study false memory in humans.
- Henry L. Roediger III and Kathleen McDermott extended the line of research in 1995 that the paradigm became popular.

- The procedure typically involves the oral presentation of a list of **related words** and requires the subject to remember as many words from the list as possible.
- An exception is statements that are pragmatically striking, such as those that require a response from us or flout the normal conventions of everyday discourse.
- These inferences are drawn in the process of comprehension.
- All of these considerations suggest that a linguistically based representational system is a poor candidate for a model of sentence memory.
- The exact **linguistic form** is not well retained
- Nonlinguistic information may **play a major role** in the retention process.
- A proposition is an idea unit; it is a statement that expresses **a factual claim**; it is the basic unit involved in the understanding and retention of text.
- Propositions correspond roughly to **verbs, adjectives, adverbs, prepositions, and subordinating conjunctions** (not nouns or pronouns).
- Proposition density is an important factor in reading comprehension.
- Sentences in print often have a complex, embedded syntax that places demands on the reader's working memory.
- The combination of text comprehension and retention, and demands on the reader's working memory suggest that proposition density might be useful in the selection of college textbooks.
- Readability formulas utilized in reading comprehension research do not estimate proposition **density.**
- Readability formulas include **Flesch-Kincaid Grade Level or Reading Ease, Degrees of Reading Power, Lexiles, and Coh-Metrix.**
- single metrics are based on length of words and sentences (Flesch Kincaid),
- readers' performance on a cloze procedure (Degrees of Reading Power and Lexiles),
- and on various language-discourse levels (Coh-Metrix).

- Sentence and clause types and nonfinite verbals are important in this research because they are directly related to **complex, embedded syntax**.

Lesson-10

- Comprehension of connected discourse depends less on the meanings of the individual sentences than on their arrangement.
- Indeed, it is entirely possible for a group of meaningful sentences to be thrown together in a way that makes no sense at all.
- In structural approaches, the aim is to discover the rules which, if followed, result in an acceptable or well-formed text.
- A central concept is the notion of cohesion.
- Halliday and Hasan (1976) define cohesion.
- The range of possibilities that exist for linking something with what has gone before is Cohesion.
- Reference deals with the links between words and objects or events in the world.
- In discourse, reference deals with the links between words (or phrases) and other words (or phrases) in discourse.
- Reference is a semantic relation whereby information needed for the interpretation of one item is found elsewhere in the text.
- We often use pronouns such as she, he, it, his, her, and their to refer to earlier items.
- **Given information** refers to information that an author or speaker assumes the reader or listener already knows.
- **new information** is the information that the comprehender is assumed to not know.
- **Direct matching:** The simplest case is surely that in which the given information in the target sentence.
- In comprehending the target sentence, we first divide it into given and new information.
- **Bridging:** In some cases, we do not have a direct antecedent for the given information but can still tie the sentences together:

- Haviland and Clark (1974) have shown that target sentences that require bridges take longer to comprehend than those for which there is a direct match of antecedents.
- In general, we form bridges when we believe the author intends for us to find a relationship between the context and the target.
- Daneman and Carpenter developed a complex reading span task to examine this trade-off.
- The task began with only two sentences in a series and progressed until a person could not recall the final words in each sentence.
- Daneman and Carpenter found a significant correlation between reading span and reading comprehension.
- When we encounter unfamiliar passages, it is more difficult to draw appropriate inferences.
- when we have information in permanent memory that helps us interpret the information, it is easier to draw inferences.

Lesson 11

- Preserved working memory is pivotal for language processing effectiveness, being an essential tool for resolving structural and lexical ambiguity during the discourse processing and comprehension.
- Based on this premise, several investigators have suggested that changes in the capacity of working memory will have an impact on the linguistic processing, leading to problems in comprehension of larger units like discourse.
- One early study that suggested that surface representations of discourse are very short-lived presented individuals with a long oral passage that was interrupted at irregular intervals.
- Individuals were asked at each interruption to write down in verbatim form as much of the preceding discourse as they could.
- Two versions of the passage were created(1) The confidence of Kofach was not unfounded. (2) Kofach had been persuaded by the international to stack the meeting for McDonald.
- The union had even brought in outsiders.
- Although the final clauses in (1) and (2) were identical, the material immediately preceding either came from the same sentence (1) or the earlier sentence (2).
- It was found that the percentage of correct recall of the next-to-last clause was far better when it was part of the current sentence than when it was part of an earlier sentence.
- These and similar results have been taken as evidence that the surface or verbatim form of a sentence is stored in working memory only until its meaning is understood, then purged to make room for the next sentence.

- There is an exception to this rule, however.
- Subsequent results indicate that we sometimes remember the exact wording over a long time period.
- Evidence for the psychological reality of propositions comes from **Kintsch and Keenan**, who showed that the number of propositions influences the time required to read a passage when preparing to recall it.
- (3) Cleopatra's downfall lay in her foolish trust in the fickle political figures of the Roman world. (4) Romulus, the legendary founder of Rome, took the women of the Sabine by force. (read these sentences for your own understanding)
- However, sentence (3) is more complex propositionally than (4), which contains four propositions.
- **Kintsch and** Keenan found that a proposition added about 1.5 seconds to the reading time.
- Later studies provide somewhat lower estimates of the time needed to encode a single proposition but support the general conclusion that the number of propositions is related to reading time.
- Further work explored the notion that discourse is stored as a network of propositions.
- From a communication standpoint, an inference is a proposition in the underlying discourse structure that is intended but not explicitly expressed by the author and thus must be drawn by the reader.
- Furthermore, evidence indicates that when we draw inferences from a text, we store the implicit propositions right alongside the explicit propositions we have derived from the text itself.
- Situational models represent the state of affairs that a text refers to.
- That is, the assumption is that as we comprehend the propositions of a text, we construct a mental or situational model of the world as described by the text.
- One possibility is a spatial layout, other is situational models.
- **Kinds of situational models**: All of the preceding examples of situational models are spatial models, but there are other kinds of situational models.
- **Zwann and Radvansky** identified a number of different types of models other than spatial models.
- One is a **causal model**, **in which the parts of a text are connected by causal relations**.
- The highest level of representation is the situation model which represents ideas of the text in an abstract format (e.g., the people, the actions, the setting, the events, and the inferences).
- Information in the situation model interacts with preexisting knowledge (i.e., episodic and semantic memory) of the reader, so that situation models can be updated online as one continues reading/listening to the text.
- As we have seen, we form surface, propositional, and situational representations during the course of comprehending discourse.
- Most of the studies we have discussed to this point have attempted to isolate one of these levels or to distinguish between different levels.
- It is also helpful, however, to set up a study that attempts to investigate how each of the levels operates in the same experiment.
- **Fletcher and Chrysler** have reported such a study.

Lesson 12

☆A schema (plural: schemata) is a structure in semantic memory that specifies the general or expected arrangement of a body of information.

☆when college students were given stories that were inconsistent with their schemata, recall was usually distorted in the direction of the schemat.

☆Bartlett's notion of a schema, although appealing, was rather vague, and modern extensions of his work have focused primarily on two issues.

☆ Genres are important because they provide us with general expectations regarding the way information in a discourse will be arranged.

☆If space is not available for the entire article as written, the editor typically deletes paragraphs near the end of the story

☆. One genre that has been studied a great deal in discourse research has been narrative discourse.

☆Narrative discourse can be contrasted with expository discourse in which the goal of the writer is not to tell a story but rather to convey information about the subject matter.

☆ In the remainder of this section, we will explore how we comprehend and experience narrative discourse.

☆Bartlett's ideas have been formalized by contemporary researchers into the concept of a story grammar.

☆In general, story grammars view narratives as consisting of a setting, one or more episodes, and then an ending.

☆Black and Bower showed that the length of one episode does not influence the recall of another.

☆Glenn reported that the episodic structure of recall is unaffected by the length of the episodes.

☆In Uncle Lew's story, a house burned down on a cold winter night and the little children inside ran barefoot into the snow of 1906.

☆Meyer, Brandt, and Bluth (1980) found that when the key points of a passage are signalled explicitly, performance improves.

☆Reder and Anderson found that retention was better when the material was presented in a condensed version rather than in a standard textbook version.

☆In a similar vein, Giora (1993) found that analogies in text did not facilitate comprehension and may actually impair recall.

Lesson-13

- The production of signs is important theoretically because it gives us an opportunity to disentangle the cognitive processes involved in translating thought into language from the physical characteristics of our speech apparatus.
- Speech shares the vocal channel with **respiration**;
- In contrast, sign production can occur entirely in parallel with, and unimpeded by, **respiratory activity**.
- Consideration of sign production in comparison with speech production can yield insights into some of the biological limits on linguistic form **(Bellugi & Studdert-Kennedy, 1980)**.
- One striking similarity is that errors occur in signing that strongly resemble those found with.
- Studies of sign language production are valuable because they enable us to **distinguish between those aspects of production that are constrained by broad biological forces and those that are specific to speech**.
- Sign language, because it exists in an entirely different mode from speech, might well differ substantially from speech in terms of grammatical organization.
- In contrast, basic similarities have been found in the two modes' organization of basic units into words or signs and in the syntactic rules by which words and signs are combined to form sentences
- similarities are illustrated by **slips of the hand**, which, like those of speech, typically involve a systematic error in a single **linguistic** unit.
- These results provide evidence that the parameters underlying signs are planned independently of one another.
- Principally, there are **two types of sign language**:
- One that relates to ordinary speech-based language and one that is independent of ordinary language.
- Speech-based sign languages represent spoken words (or their spelling) and the order of these words or morphemes as they appear in ordinary spoken languages, such as **Swedish, English, and French**.
- This contrasts greatly with such sign languages as American Sign Language and British Sign Language, which are **not speech based** and not mutually intelligible.
- These sign languages are independent of the ordinary spoken language, having developed their own words and grammatical systems for the production and understanding of sentences.
- We shall call these Independent Sign Languages (ISLs).
- Sign language based on the speech of ordinary language can be of **two different** kinds: one that represents the morphemes of speech and one that represents spelling (orthography).

- Let us begin with the latter, which is easier to explain and which, by the way, every signer of whatever system must learn so as to be able to express proper nouns such as the names of people or places.
- Four basic grammatical concepts are duality of patterning, morphology, phrase structure, and linguistic productivity.
- Words are composed of phonemes, which, in turn, have distinctive features.
- In each instance, the smaller units are combined in a rule-governed manner to produce the larger units.
- Words consist of one or more units of meaning or morphemes.
- The system of **grammatical** morphemes in a language provides speakers with a way of signaling subtle differences in meaning.
- Phrase-structure rules codify our intuitions about the groupings of words in a sentence.
- Some sentences are ambiguous and may be grouped in more than one way.
- Linguistic productivity refers to the fact that there is no limit to the number of sentences in a language.
- One type of phrase-structure rule, that of recursion, is responsible for some of this productivity.
- Unlike speech, signs are expressed in **visual or spatial form**.
- This enables us to examine the extent to which the grammatical concepts we have just considered generalize to language in a visual modality.
- American Sign Language is sharply distinguished from manual forms of English that translate English sounds into signs.
- The best known is fingerspelling, which, as the name implies, translates English words letter by letter into manual form.
- It is a secondary gestural system, derived from the English language.
- In contrast, ASL is independent of English and derived from French Sign Language.
- Although in the past ASL was regarded as **mere pantomime or grammatically deficient** in various ways, several decades of scholarly research on ASL have put these ideas to rest.
- Even if we accept the notion that ASL is an autonomous language, we must ask what its relation to spoken languages is.
- **Evaluation of Grammars213**

- **Chomsky** has suggested **three** criteria.
- **Observational adequacy**: **First, the grammar must specify what is and what not an acceptable sequence in the language is**.
- This criterion, referred to as observational adequacy, applies at several levels of language.
- We know at the phonological level that pbort is not an acceptable sequence.
- Similarly, at the syntactic level we want the grammar to have rules that generate grammatical sentences without also generating strings of words we would regard as ungrammatical.
- A grammar is observationally adequate if it generates all of the acceptable sequences in a language and none of the unacceptable sequences.
- **Descriptive adequacy**: **The second criterion is that the grammar must specify the relationships between various sequences in the language, a criterion known as descriptive adequacy**.
- It is not enough for the grammar to mark a sequence as permissible; it must also explain how it relates to other sentences that are similar in meaning, opposite in meaning, and so on.

- If, for example, two sentences are similar in meaning but differ in syntax, the grammar should be able to explain this fact.
- **Explanatory adequacy**: The third criterion is called explanatory adequacy. **Chomsky** points out that it is theoretically possible for a **number of grammars**, all based on different principles, to attain these two forms of adequacy.
- Chomsky's answer pertains to language acquisition in children.
- He suggests that the child learning a language is presented with samples of the language and must determine the grammar from these samples.
- Chomsky notes, however, that even though the incoming data may be consistent with any number of grammars, children choose one particular grammar.
- This implies that certain innate language constraints enable the child to deduce the correct grammar.
- These innate language mechanisms would presumably be related to linguistic universals common to all languages.
- Thus, the final level of adequacy goes beyond the ability to describe patterns in a particular language; instead, it involves the ability to explain the role of linguistic universals in language acquisition.
- Psychological reality of grammar: As indicated earlier, much psycholinguistic research in the **early and mid-1960s** was based on transformational grammar.
- This research was guided by the belief that the structures and rules of transformational grammar were psychologically real; that is, they were a part of how people comprehend and produce language.
- One assumption that was made was that the surface structure was the starting point for comprehension and that the deep structure was the end point; the roles were assumed to be reversed for production.
- If so, then it would be reasonable to assume that the distance between surface and deep structure (as measured by the number of transformations in a sentence's derivation) would be an accurate index of the psychological complexity of the sentence.
- This view was called the of **derivational theory complexity** or DTC. Early studies were encouraging.
- A variety of studies showed that negative sentences such as
- The sun is not shining. were more difficult to comprehend than the corresponding affirmative form such as (2) The sun is shining.
- But these sentences differ in meaning as well as transformational complexity, so this point is hardly conclusive.
- Later studies directly contradicted **DTC**. Sentence (3) is, for example, transformationally more complex than (4):
- (3) The boy was bitten. (4) The boy was bitten by the wolf. In transformational theory, (3) requires a transformation that deletes the phrase by the wolf, so DTC would predict it would be more difficult to comprehend than (4).
- However, neither intuition nor experiment has revealed any relationship to processing difficulty.
- Similarly, there is no psychological difference between sentences that have undergone particle-movement transformation and those that have not. These studies have been reviewed extensively elsewhere.

- As **Berwick and Weinberg (1983)** pointed out, however, these results do not necessarily mean that the linguistic theory of transformational grammar is faulty.
- It could be that the linguistic theory is correct but that some of the psychological assumptions guiding DTC are faulty.
- More recent work has been more favorable to the hypothesis that linguistic theory has psychological reality.
- Consider this sentence: (5) The dentist from the new medical center in town was invited by the actress to go to the party. The use of the passive voice results in the movement of the NP that is the object of the verb (dentist) from the object position to the subject position.
- According to recent grammatical theory, it is assumed that the moved constituent leaves a trace at its earlier location. Thus, the presumed linguistic representation of (5) would be more like (6)The dentist from the new medical center in town was invited [trace] by the actress to go to the party.
- If this proposal has psychological reality, then the hypothesis would be that comprehenders would be likely to reactivate the moved noun (dentist) when its trace was encountered.
- **Osterhout and Swinney** (1993) have provided evidence that comprehenders do this.
- Participants responded rapidly when words semantically related to the moved noun were presented in the trace position.
- It is as if they were thinking about dentist which made it easier to respond to a semantically related word, such as tooth.
- Responses were slower either before or after the trace position.
- **The centrality of syntax**: There have long been controversies within linguistics regarding the proper way to characterize linguistic knowledge.
- As we have seen, phrase-structure rules are insufficient in themselves to account for our linguistic capacities, and these insufficiencies led Chomsky to propose transformational grammar.
- In the years since transformational grammar was formulated, it has gone through a number of changes.
- In the most recent version, Chomsky (1995) has eliminated many of the transformational rules in previous versions of grammar and replaced them with broader rules, such as a rule that moves one constituent from one location to another. It was just this kind of rule on which the trace studies were based.
- Although newer versions of the theory differ in several respects from the original, at a deeper level they share the idea that syntactic structure is at the heart of our linguistic knowledge.
- This view has been controversial within linguistics.
- We will discuss two alternative linguistic theories. One alternative approach is supplied by lexical theories of grammar.
- In lexical theories (for example, Bresnan, 1978), greater emphasis is placed on individual lexical items (words) than is given in more structural theories, such as transformational grammar.
- This view has been influential in recent years in diverse areas of psycholinguistics, including **language comprehension, language production, and language development**.
- In most grammars, the lexical entry for a word includes its meaning, its spelling, its pronunciation, and syntactic characteristics such as part of speech.

- **Bresnan's lexical-functional grammar**, lexical entries also include the various forms of the word (for example, kiss, kissed, kissing) and the different kinds of sentences into which each form would fit.
- For verbs, this includes the arguments or semantic roles, such as the agent (the person doing the action) and the patient (the one to whom the action is done) that are associated with the verb, as well as the surface structure designation, such as subject or object, that goes with it.
- The lexical entry for kiss would indicate its underlying semantic structure as kiss: (agent, patient)
- The verb requires both an agent and a patient (Mother kissed is not a grammatical sentence).
- In addition, the entry includes various forms of the word, including kiss: agent = subject, patient = object and (be) kiss: agent = object; patient = subject.
- The first verb form, used in sentences in the active voice, assigns the agent role to the **surface-structure** subject and the patient to the surface object.
- The second form, used in passive sentences, assigns the patient to the subject and the agent to the object of the preposition by.

Lesson-14

- **Planning / preparation**: Language production consists of several interdependent processes which transform a nonlinguistic message into a spoken, signed, or written linguistic signal.
- Though the following steps proceed in this approximate order, there is plenty of interaction and communication between them.
- The process of message planning is an active area of psycholinguistic research, but researchers have found that it is an ongoing process throughout language production.
- Research suggests that messages are planned in roughly the same order that they are in an utterance.
- After identifying a message, or part of a message, to be linguistically encoded, a speaker must select the individual words— also known **as lexical items**—to represent that message. This process is called lexical selection.
- The words are selected based on their meaning which in linguistics is called **semantic information.** Lexical selection activates the word's lemma which contains both semantic and grammatical information about the word.
- The first important goal of conceptual preparation is to establish which parts of the conceptually available information are going to be encoded and in what order.
- The second goal is to convert the conceptual information into a format that is a suitable for the linguistic formulation processes.
- One important open issue concerning conceptual preparation is how to characterize the mapping between conceptual and grammatical encoding.
- First is the question of whether conceptual preparation takes language specific properties into account. Languages differ in which conceptual or formal properties need to be realized as a detail of the sentential form.
- For example, in English the word 'friend' does not carry information concerning the sex of the friend.
- In Spanish, the corresponding (friend) word is differentially inflected for a **man** ('amigo') or a **woman** ('amiga').

- In English, adjectives used as predicates (e.g., 'tall' in 'The friend of Luis is tall') do not agree in gender with the noun;
- In Spanish they do (e.g., 'El amigo de Luis es alto' or 'La amiga de Luis es alta'). Thus, in these two languages, conceptual information concerning natural gender must (Spanish) or need not (English) be conveyed by a sentence.
- **Grammatical information**: Critical grammatical information includes characteristics such as the **word's syntactic category (noun, verb**, etc.), what objects it takes, and grammatical gender if it is present in the language.
- Using some of these characteristics as well as information about the thematic roles of each word in the intended message, each **word** is then assigned the grammatical and thematic role it will have in the sentence.
- **Function morphemes,** like the plural /s/ or the past tense /ɪd/, are added in this stage as well.
- **Grammatical encoding** refers to the processes involved in developing a syntactically well-formed sentence.
- It comprises first those processes that map the relationships among the participants in a conceptual representation (e.g., agent, patient, etc.) onto functional syntactic relations between the words of a sentence (e.g., subject, direct object, etc.).
- Next, on the basis of the resulting hierarchically organized syntactic frame for the sentence, the words in the sentence are linearized in a manner allowed by the language being spoken.
- The first step is also referred to as functional level processing, the second as positional level processing.
- Distinguishing between **building hierarchical and linear frames** provides a solution to an important problem that the language production system faces.
- Speech production has to be incremental to allow for fluent utterances, but at the same time, the resulting utterance has to obey language specific constraints that force the use of only certain word orders.
- Assuming incremental conceptualization, however, the order in which parts of a conceptual message are processed do not necessarily correspond to a word order allowed in the speaker's language.
- This problem can be solved by separating the construction of hierarchical structures from the serial ordering of the words.
- In this way, hierarchical structures can be built in an incremental manner as soon as lexical elements are available; these can then be mapped to permissible linearly ordered positions.
- Evidence compatible with such a separation comes from studies showing that the linear position of words can be primed by the previous presentation of the same linear order, even if the hierarchical structure differs.
- **Phonological encoding**: After an utterance, or part of one, has been formed, it then goes through phonological encoding.
- In this stage of language production, the mental representation of the words to be spoken is transformed into a sequence of speech sounds to be pronounced.
- The speech sounds are assembled in the order they are to be produced.
- Phonological encoding refers to the processes that are responsible for determining the phonological word forms and prosodic content of the sentence.

- First, the phonemes of words are retrieved from the mental lexicon, together with a metrical frame which specifies stress pattern and number of syllables in the word.
- Following this retrieval process, the resulting sequence of phonemes is syllabified according to a language specific set of syllabification rules.
- The domain of syllabification is assumed to be the phonological word which can but need not coincide with a lexical word.
- It should be noted that in this view, the syllabic structure of an utterance is computed on-line.
- The reason for this assumption is that the actual syllabification of a word in running speech depends on the context in which it appears. For example, the word 'deceive' in isolation is syllabified as 'de-ceive,' but in the context of the utterance 'deceive us,' the syllable structure becomes 'de-cei-veus.'
- This observation also provides a functional reason as to why the phonological form of a word is not stored and retrieved as one entity: such a representation would have to be broken up into its constituent parts whenever the stored syllabification of a word does not agree with its syllabification in the context of running speech.
- The representation formed by phonological encoding, a syllabified phonological code, forms the input for the articulatory processes which realize this code as overt speech.
- It is an open issue whether the transition from phonological encoding to articulation involves accessing a 'syllabary,' i.e., memory representations specifying the motor tasks that have to be performed to generate each syllable.
- Slips-of-the-tongue are speech errors in which intended utterances are rearranged between other words or sounds.
- According to psychologist Gary Dell, slips-of-the tongue are significant because they show a person's widespread knowledge about language, including its sounds, structures, and meanings.
- There are three types of slip-of-the-tongue errors.
- These types include **sound errors, morpheme errors, and word errors**.
- A sound error occurs when the sounds in words close by are exchanged. For example, instead of saying "flower pot," one says "**power flot**."
- A morpheme error occurs when morphemes, which are the smallest meaningful units in language, are switched in words close by. For example, instead of saying "self-destruct instruction," one says, "**self-instruct destruction**."
- Word errors occur when actual words are rearranged. For example instead of saying, "reading a book to my dog," one says, "reading a dog to my book."
- Errors in speech production and perception are also called performance errors.
- According to Sigmund Freud, slips-of-the-tongue reveal the thoughts and desires of the unconscious mind.
- These slips-of-the-tongue are called Freudian slips or parapraxes.
- Speech errors are made on an occasional basis by all speakers.
- They occur more often when speakers are nervous, tired, anxious, or intoxicated.
- During live broadcasts on TV or on the radio, for example, nonprofessional speakers and even hosts often make speech errors because they are under stress.
- Some speakers seem to be more prone to speech errors than others. For example, there is a certain connection between stuttering and speech errors.

- Charles F. Hockett explained that "whenever a speaker feels some anxiety about possible lapse, he will be led to focus attention more than normally on what he has just said and on what he is just about to say.
- These are ideal breeding grounds for stuttering. Slips of the tongue are a normal and common occurrence.
- One study shows that most people can make up to as much as **22 slips** of the tongue **per day**.
- Speech errors are common among children, who have yet to refine their speech, and can frequently continue into adulthood.
- When errors continue past the age of **9**, they are referred to as "**residual speech errors" or RSEs**.
- They sometimes lead to embarrassment and betrayal of the speaker's regional or ethnic origins.
- It is also common for them to enter the popular culture as a kind of linguistic "flavoring."
- Speech errors may be used intentionally for humorous effect, as with Spoonerisms.
- Although speech errors cover a wide range of semantic content, there appear to be only a small number of basic types. Examples of the eight types are given in Table 8.1, with the words that were apparently intended in parentheses.

Major **types of slips** of the tongue

Types	Examples
Shift	That's so she'll be ready in case she decide to hits it (decides to hit it)
Exchange	Fancy getting your model renosed (getting your nose remodeled)
Anticipation	Bake my bike(take my bike)
Perseveration	He pulled a pantrum (tantrum)
Addition	I didn't explain this clarefully enough (carefully enough)
Deletion	I'll just get up and mutter intelligibly (unintelligibly)
Substitution	At low speed it's too light (heavy)
Blend	That child is looking to be spaddled (spankedy/paddled)

Garrett has identified four generalizations about speech errors that reappear with striking regularity. First, elements that interact with one another tend to come from similar linguistic environments, as indicated by examples (2) through (4): (2) The little burst of beaden (beast of burden). (3) You're not a poojin pitter-downer, are you? (pigeon putterdowner) (4) Children interfere with your nife lite (night life).

- Notice that the phonetic segments in the beginning of a word tend to be exchanged with other initial segments; the same is true for middle and final segments.
- Moreover, exchanges of segments are more common when the segments that precede them are similar. The exchange of **/f/ and /t/** in sentence (4) follows this principle. Second, elements that interact with one another tend to be similar to one another. In particular, consonants are invariably exchanged or shifted with other consonants but not with vowels. Errors involving similar sounds, such as in sentence (5), often have little relation to meaning but are based, instead, on phonetic similarity:
- Speech errors, the bane of performers on live television and radio, are systematic and typically fall into one of eight categories: exchanges, substitutions, additions, deletions, anticipations, perseverations, blends, and shifts.
- Various hypotheses concerning the basis for such errors have been advanced.
- One of the more prominent has been Freud's view that errors occur because we have more than a single plan for production and that one such plan competes with and dominates the other.
- Although a Freudian type of explanation may apply to some speech errors, more recent thinking has focused on the psycholinguistic processes underlying speech errors.
- The most common interpretation is that we produce speech through a series of separate stages, each devoted to a single level of linguistic analysis.
- Errors typically occur at one level, but not others, during the production process.
- The Freudian explanation: One intriguing idea is that speakers have more than one idea in mind at a time.
- During the 1992 campaign, President George Bush began his remarks for one speech by saying (6)I don't want to run the risk of ruining what is a lovely recession (reception).
- This, of course, could be construed as simply a sound error, as the two words are similar phonologically. But it could also be evidence that the president was pre-occupied with the recession (and its effect on his campaign).
- Freud emphasized the role of psychodynamic factors in making certain types of content more available than others.
- He argued that these errors 'arise from the concurrent action—or perhaps rather, the mutual opposing action—of two different intentions' **(Freud**, 1916–1917/1963, p. 44).
- One of these actions was thought to constitute the conscious intention of the speaker, whereas the other pertained to a more disturbing thought or intention that interfered with the former.
- Sometimes, the disturbing comment would be censored; but, on other occasions, the outcome of this hypothetical intrapsychic conflict would be a slip of the tongue that expressed some aspects of the less conscious intention.
- Examples consistent with Freud's position include a general who referred to a group of injured soldiers as battled **scared (scarred**) and a speaker extolling the achievements of a fellow worker who had just **expired (retired**).
- Freud's position was that virtually all speech errors were caused by the intrusion of repressed ideas from the unconscious into one's conscious speech output.
- Although the Freudian interpretation may be appealing in cases in which the slip of the tongue results in a word with emotional significance, many slips seem to reflect simpler processes, such as anticipation (a meal mystery instead of a real mystery) and perseveration (he pulled a pantrum in place of he pulled a tantrum) of phonetic segments.

- In these latter cases, it seems to be unnecessarily complicated and unconvincing to claim that the error originated from intrapsychic conflicts.
- A Psycholinguistic explanation: Most recent psycholinguistic and linguistic thinking has focused on the insights gained in understanding language mechanisms (not unconscious motivations) from the study of speech errors.
- In this respect, errors of linguistic performance occupy a role in psycholinguistic theories similar to that played by aphasic disorders.
- The types of language breakdowns that occur in each case provide important insights for normal language functioning.
- For example, a study has shown that many of the segments that change and move in speech errors are precisely those postulated by linguistic theories, lending support to the notion that linguistic units such as phonetic features, phonemes, and morphemes constitute planning units during the production of an utterance.
- One view of language production is that we produce utterances by a series of stages, each devoted to a different level of linguistic analysis.

Lesson-15

- The **pioneering studies** of Fromkin suggested that the process of planning speech can be viewed as a series of stages, each devoted to one level of linguistic planning.
- **Fromkins Model of speech production:**
  - **Stage 1: identification of meaning**- a meaning to be conveyed is generated
  - **Stage 2:** selection of a syntactic structure- a syntactic outline of the sentence is constructed, with word of slots specified.
  - **Stage 3:** generation of intonation contour- the stress values of different word slots are assigned.
  - **Stage 4:** insertion of content words- appropriate nouns, verbs and adjectives ar retrieved from the lexicon and placed into word slots.
  - **Stage 5**: formation of affixes and function words- function words (articles, conjugations, preposition), prefixes and suffixes are added.
  - **Stage 6:** specification of phonetics segments- the sentence is expressed in terms of phonetic segments, according to phonological rules.
- What evidence can be given that the stages **hypothesized in** above are actually independent of one another?
- Probably the clearest evidence is that the vast majority of speech errors contain mistakes at only **one level** of planning.
- One of Fromkin's examples is **sentence (1),** which was pronounced so-er: **(1): Singing sewer machine (Singer sewing machine):** Here the error is at stage 5, as the suffixes are exchanged for

one another. Yet the rest of the utterance—the content words, stress values, and syntactic structure— remained unaltered.

- An even more **striking** example of the point is Garrett's **sentence (2):** (2) Stop beating your brick against a head wall. (Stop beating your head against a brick wall.):
- In addition to the **stages of planning**, some intriguing evidence indicates that **editing processes** intervene between the planning of an utterance and its articulation.
- These editing operations might provide a last check to determine whether the **planned utterance** is linguistically and socially acceptable.
- It is clear that some **monitoring and editing** processes occur after a speech segment is uttered; after all, we often spontaneously correct ourselves. The question we want to consider now is whether we also have editing processes prior to articulation.
- One intriguing idea is that speakers have more than one idea in mind at a time. During the 1992 campaign, President George Bush began his remarks for one speech by saying (3): statement (3) I don't want to run the risk of ruining what is a lovely recession **(reception**).
    - This, of course, could be construed as simply a sound error, as the **two** words are similar phonologically. But it could also be **evidence** that the president was preoccupied with the recession (and its effect on his campaign).
- Or consider a student who explains that he wants to postpone an exam with statement (4): **statement 4: Last night my grandmother lied (died).**
    - This could be an innocent phonological error, but then again, the slip could reveal the student's thinking more than he wishes.
- Freud emphasized the role of psychodynamic factors in making certain types of content more available than others.
- Freud argued that these errors 'arise from the concurrent action—or perhaps rather, the **mutual opposing action**—of two different intentions' (Freud, 1916–1917/1963, p. 44).
- One of these actions was thought to constitute the conscious intention of the speaker, whereas the other pertained to a more disturbing thought or intention that interfered with the former.
- Sometimes, the disturbing comment would be censored; but, on other occasions, the outcome of this **hypothetical intrapsychic conflict** would be a slip of the tongue that expressed some aspects of the less conscious intention.
- Examples **consistent with Freud's position** include a general who referred to a group of **injured soldiers as battled scared** (scarred) and a speaker extolling the achievements of a fellow worker who had just expired (retired) (Ellis, 1980).
- Freud's position was that virtually all speech errors were caused by **the intrusion of repressed ideas f**rom the unconscious into one's conscious speech output.

- Although the **Freudian interpretation** may be appealing in cases in which the **slip of the tongue** results in a word with emotional significance, many slips seem to reflect **simpler processes,** such as anticipation (a meal mystery instead of a real mystery) and perseveration (he pulled a pantrum in place of he pulled a **tantrum**) of phonetic segments.
- In these latter cases, it seems to be unnecessarily complicated and **unconvincing** to claim that the error originated from intrapsychic conflicts. Still, these more common speech errors demand an explanation.
- Speech errors from both spontaneous speech as well as laboratory studies have provided researchers with a body of data about the **production** of language.
- Theories of how we proceed from message to **linguistic structure** come in two types.
- **Serial models** assume that we begin with the overall idea of an utterance, followed by syntactic organization, content words, morphemes, and phonology.
- **Slips of the tongue** typically involve just one level of planning, with other levels unaffected.
- There may be a final stage, after the planning of an utterance but before its articulation, that edits the utterance-to-be in a manner not inconsistent with **Freud's ideas.**
- Recent alternatives to the stage models are parallel models of production.
- These models assume that the linguistic message is organized at **semantic, syntactic, morphological, and phonological levels**.
- **Activation of a node** at one level may trigger activation of nodes at other levels, and feedback may occur from **morphological and phonological levels** back to higher levels of processing.
- **Models** organized along these lines have been shown to account for several important research findings.
- A line of research that may be helpful in evaluating serial and parallel models concerns **number agreement.**
- In English, in order for a sentence to be **grammatical** there needs to be number agreement between subjects and either verbs or pronouns. The concerts this Summer have been wonderful, not The concerts this Summer has been wonderful, and The pitcher's fastball is his best pitch, not
- The pitcher's fastball is their best pitch. We sometimes make **agreement errors** that are **instructive**. For example, in sentence (5), the head noun (time) controls the correct form of the **subsequent** verb (is), but we sometimes err by using a form of the **verb** (are) that matches the immediately preceding word (games). (5) For example, the time for fun and games is over.

Lesson-16

- Once we have organized our thoughts into a linguistic plan, this information must be sent from the brain to the muscles in the **speech system** so that they can then execute the required movements and produce the desired sounds.

- However, it is useful to understand certain basic aspects of articulation, in anticipation of our later comparison of the **production** of **signed versus spoken** language.
- Three Systems of Muscles Fluent articulation of speech requires the coordinated use of a large number of muscles. These muscles are distributed over three systems: the **respiratory, the laryngeal, and the supralaryngeal or vocal tract.**
- Several studies have converged on the conclusion that we alternate between planning speech and implementing our plans.
- Consider first a study performed by **Henderson, Goldman-Eisler, and Skarbek (1966),** who analyzed the hesitations and fluent speech of individuals being interviewed.
- Note that there appears to be an alternation of steep parts (**primarily pausing**) and flat parts **(mainly speech)**. Henderson and his colleagues found that all of the participants showed this **cycle of hesitation and fluency**, although the ratio of speech to silence varied among speakers.
- Speakers routinely monitor their utterances to ensure that they are saying what they wanted to and in the way they wanted to.
- When errors are detected, speakers interrupt their speech nearly **immediately and begin editing** their utterance.
- Both the use of **editing expressions and the linguistic structure** of the repair itself appear to facilitate listener comprehension.
- It appears that the editing expression conveys to the listener the kind of trouble that the **speaker** is correcting.
- James (1972) analyzed utterances containing expressions such as uh and oh, suggesting that these convey different meanings.
- For instance, in sentence (1), the uh suggests that the speaker paused to try to remember the exact number of people. In contrast, sentence (2) would be used when the speaker **did not know** the precise number but was trying to choose a number that was approximately correct. (1) I saw ... uh ... 12 people at the party. (2) I saw ... oh ... 12 people at the party.
- We now consider some of the linguistic properties of American Sign Language (ASL). Unlike speech, signs are expressed in visual or spatial form.
- This enables us to examine the extent to which the grammatical concepts we have just considered generalize to language in a visual modality.
- American Sign Language is sharply distinguished from manual forms of English that translate English sounds into signs.
- The best known is fingerspelling, which, as the name implies, translates English words letter by letter into manual form. It is a secondary gestural system, derived from the English language.

- In **contrast,** ASL is independent of English and derived from **French** Sign Language. Although in the past ASL was regarded as mere pantomime or grammatically deficient in various ways, several decades of scholarly research on ASL have put these ideas to rest.
- Even if we accept the notion that ASL is an autonomous language, we must ask what its relation to spoken languages is. We will begin to answer this question by considering some of the differences between signed (especially ASL) and spoken languages and then some of the similarities.
- **American Sign Language** has its own set of grammatical rules and is a language that is independent of English.
- Our preliminary look at ASL indicates some striking similarities in its **grammatical** organization, suggesting that some of the basic concepts we have been discussing might be **universal.**
- At the same time, there are significant **differences** between ASL and English, and we will examine these further. Because the similarities and differences between ASL and spoken languages are so intriguing, we will return periodically to the study of ASL throughout this book.
- Studies of production rate reveal differences between the two modes. Speakers achieve differences in speech rate primarily by varying the number of pauses, whereas signers vary the duration of **signed segments** and both the

Lesson-17

- The linguist Charles Fillmore has stated that the language of face-to-face conversation is the basic and primary use of language.
- Debates, for example, typically have topics specified in advance, and rules specifying who can speak at a given time and for how long are also usually agreed on ahead of time.
- Speakers typically speak for an extended period of time.
- Ceremonies, such as an awards dinner, are also formalized.
- Turns are identified rather clearly, with formal introductions given for each speaker.
- Meetings are typically less formal than either ceremonies or debates.
- It is not uncommon for one member of a committee to dominate the proceedings.
- Finally, conversations are the least formal of these four types of oral discourse.
- Yet, in the absence of formal rules, we have implicit communicative conventions that help organize everyday conversations.
- Theoretically the number of possibilities for opening conversations is infinite; in practice we do so in a limited number of ways.
- This quickly establishes the alternation of turns that is central to conversation:
- A asks a question, B replies, followed by a sequence of the form ABABAB.
- Conventions are also at work when we close conversations.
- Schegloff and Sacks (1973) suggest that one way to end a conversation is to present a preclosing statement like we-ell, so-o-o, or OK, which signals a readiness to end the conversation.
- The listener then may accept the statement with an utterance such as yeah or OK.
- Albert and Kessler (1978) list several ways in which we end conversations, including summarizing the content of the conversation, justifying ending contact at this time (I have another meeting),

expressing pleasure about each other, making reference to the ongoing relationship and planning for future contact (see you later), and wishing each other well (take care, have a good trip).

- Albert and Kessler propose that these closing moves form a sequence, with the items occurring in the order indicated earlier.
- By presenting one of these closing statements and having one's conversational partner reciprocate, the conversationalists are implicitly negotiating an end to the conversation.
- When they are done with a particular conversation, they simply walk away. Conversations become more complicated when more than two people are present.
- According to Sacks and colleagues (1974), turn taking during conversations operates by three implicit rules.
- The first rule states that the current speaker is allowed to select the next speaker.
- The second rule is that of self-selection: If the first rule is not used, another person may speak up.
- The third rule states that the current speaker can continue, although she or he is not obligated to do so.
- As Grice has noted, there is a strong social convention to ''be relevant.'' Schank argued that there are, indeed, rules of this kind, although it is probably more accurate to say that they govern rather than severely restrict our responses.
- This is reflected in the observation that while some responses are clearly odd, a wide range of ''acceptable'' responses to any statement is possible.
- Clark has pointed out that conversations often take place in a context in which various types of nonparticipants are also present.
- Bystanders are those who are openly present but do not participate in the conversation. Eavesdroppers are those who listen in without the speaker's awareness.
- If I am having lunch with Hal and Greg stops by, I may, after introducing the two, chat briefly with Greg in such a way as to define Hal as a side participant.
- My question and Greg's answer can be phrased in such a way that Hal is completely unaware that anything significant has been discussed.
- We resort to a variety of strategies when dealing with over hearers, including disclosure, concealment, and indifference (Clark & Schaefer, 1992).
- We resort to a variety of strategies in these kinds of situations to conceal our meaning from eavesdroppers, including referring to personal events (for example, the event we talked about yesterday) and using private codes such as in-group jargon or even foreign languages.
- When speakers address their listeners, they must also take over hearers into account.
- I have to this point sketched out a series of general principles about how conversations take place—taking turns, distinguishing participants from nonparticipants, and so on.

Lesson-18

- Common ground: One concept that is helpful here is what Clark calls common ground, which refers to the shared understanding of those involved in the conversation.
- For knowledge to qualify as common ground, person A must know a given information X, and person B must know X, and A must know that B knows, and B knows that A knows, and so on; that is, both parties are aware that they share the information.

- Some of this common ground is culturally based, such as cultural values, commonly held beliefs, or culturally prescribed roles.
- When you have a conversation with your academic adviser, your discussion is linked to these roles.
- Other types of common ground are more personal, in which shared experiences influence the nature of the conversation.
- Early studies of gender differences found that men interrupt women more than vice versa, a result that has not been found as often recently.
- In recent years, some scholars within linguistics, sociology, and psychology have examined the strategies of interpreting conversations among different participants.
- the linguistic differences—tag questions, minimal responses, and so on—do not always differentiate women and men is that couples differ in the ways that they share power.
- In turn taking, Edelsky recorded the verbal behavior of women and men only to find that there were different kinds of turns; some turns had a clear speaker while others listened or responded; other turns were more collaborative in nature, with several people sharing the turn.
- Men took longer turns during the former type, but there were fewer differences between genders in the collaborative turns.
- Conversational settings shape conversational processes.
- Friends tend to converse in different ways than do acquaintances and strangers.
- Some studies of gender differences reveal that males speak more and interrupt more than females.
- When the primary means of achieving therapeutic results is through language, we would expect that therapists are especially skilled at conversational processes.
- We may distinguish three main tasks during therapy.
- First, the therapist listens carefully as the client reports experiences, issues, and concerns.
- Second, the therapist interprets the client's experiences and symptoms.
- Third, the therapist collaborates with the client regarding potential courses of action.
- These tasks are not necesilsary organized sequentially; therapeutic sessions interweave data, interpretation, and suggestion in a complex pattern.
- Consider first the process by which the client presents experiences to the therapist.
- Just as the therapist is the authority on the process of interpreting emotional experiences, it is the client who is the authority on the experiences themselves.
- The therapist does not challenge the client's reporting of experiences, for these statements provide the "raw data" for the therapeutic session.
- In contrast, it is acceptable and sometimes useful for the therapist to challenge the client's interpretation of another person's experiences, because the therapist is the expert on the interpretation of emotions.
- Consider the example of a client (C) reporting a given event and simultaneously interpreting it and the response of the therapist (T).
- Relatively little work has been done on conversations in other institutional settings, but at least a preliminary comparison with therapeutic discourse may be attempted.
- Most institutional settings identify a particular individual (therapist, judge, academic adviser, physician, and so on) as the authority figure.

- Physicians probably occupy an intermediate position on a continuum of how strictly or loosely institutional authority is wielded.
- Like therapists, physicians require data from the patient to be of much help, and good physicians listen carefully to their patients' symptoms and concerns.
- Once the test results are in, the physician interprets their significance to the patient and either recommends a particular course of action or outlines the alternative possible actions (Parsons, 1975).
- Diagnoses may vary from a single word (for example, bronchitis) to a detailed description of a condition.
- As Heath (1992) has observed, the diagnosis is a pivotal point in the consultation between patient and physician.
- It marks the end of the ''data-gathering'' phase and begins (and in fact is the basis for) the discussion of possible treatments. And it is the province of the physician to form this medical judgment.
- If the patient offers candidate diagnoses, the physician is likely to defer consideration of them until the examination or diagnostic tests are complete.

Lesson-19

- Until the early part of their second year, infants communicate with their world primarily in nonverbal ways: they tug at people's clothes, point at desired objects, and wave bye-bye.
- These gestures, though basic, reveal a good deal about the infant's understanding of how communication works.
- It appears that the emergence of these communication skills is made possible by advances in the child's understanding of how actions can be used as means for achieving desired goals.
- These advances take place in the first year of life, suggesting that infants' understanding of communication precedes and facilitates much of the child's acquisition of phonology, syntax, and semantics.
- Despite the richness of the language infants receive in the first year of life, it is some time before they are able to speak themselves.
- Before they use language to communicate, they communicate with gestures. Well before 10 months of age, children engage in a lot of vocal behavior that appears to have some communicative value.
- Children's smiles and (most definitely) cries elicit parental behavior. Moreover, different cries are discriminated by parents, and these yield responses that differ in urgency as well as type.
- Still, these sounds are not true forms of intentional communication because infants do not display flexible, goal-directed behavior. For example, if a cry is ineffective in obtaining adult attention, young infants do not turn to another behavior, such as banging an object against the side of the crib.
- Thus, although infants' cries generally elicit parental responses, the infant is not using the cry for that purpose.
- Rather, it is simply a built-in response with predictable consequences.
- Children's acquisition of the sound system of their language does not occur in isolation of the communicative processes we have just discussed.

- Rather, children come to the task of learning phonology with some knowledge of how to communicate in nonverbal ways.
- The prelinguistic infant knows how to use gestures to make assertions and requests and, once early speech sounds are mastered, they are quickly used for these same communicative functions.
- The child's first attempts at producing sounds, however, have more to do with practicing with the sound system than with communicating with others.
- Eventually, the abilities to communicate without words and vocalize without meaning merge into aproductive and communicative speech.
- The task of identifying what the child knows about phonology is difficult, for the ways in which phonological knowledge is expressed can often be rather indirect.
- Consider again the example presented at the beginning of the chapter, in which a child named Lisa pronounces her own name as Litha but objects when an adult does the same.
- Apparently a child can perceive a distinction that she cannot produce, an occurrence that has been christened the fis phenomenon after a child who called fish fis (Berko & Brown, 1960).
- Thus, we cannot simply look at children's production to assess their perception of the phonology of their native language.
- Our survey of phonological development begins with the child's perception of speech, and then turns to the production of speech

- Children begin by focusing on words related to the here and now, an observation that fits well with Piaget's description of the sensorimotor period of cognitive development.
- Many of their early words consist of nominals that refer to concrete aspects of their environment.
- They learn the names of the toys they play with, the clothes they wear, and the food they eat.
- Children have a bias toward objects that change or move in response to their actions; they are more likely to learn the word ball than the word chair. Their early vocabulary, however, is not limited to nominals.
- As Nelson has shown, children use words from various grammatical classes early on. Nelson found that general nominals such as ball and car were most prevalent, followed by specific nominals (Mommy), action words (up, go), modifiers (dirty, pretty), personal and social words (please, want), and function words (what, for).
- Children begin to speak in word combinations by about 2 years of age, and over the course of the next few years, they make impressive advances in grasping the grammar of their native language.
- These aspects of grammar, of course, differ from language to language. Children learning English must pay close attention to word order, which is the primary way in which meaning is signaled.
- Those acquiring a more inflected language, such as Turkish, must spend a relatively greater amount of time learning the different forms or conjugations of verbs. These language differences surely play an important role in language acquisition.
- There are, however, important similarities in children's early grammatical efforts. Slobin has suggested that at least the early stages of grammatical development are similar in all of the world's languages.

- Studies have now been conducted on dozens of different types of languages, and these have found that what Slobin calls basic child grammar is a universal construction of children learning their native language.
- In this section, we will consider the structure of basic child grammar and some ideas researchers into child language have developed as to what rules comprise this grammar as well as review evidence that indicates individual differences in early language acquisition.
- The structure of early utterances: It may seem odd to talk of two-word utterances as sentences having a grammatical structure.
- After all, early utterances such as all gone baby and more crayon are hardly grammatical by adult standards and may appear to be little more than random combinations of previously acquired words.
- Most investigators of child language, however, agree with Sachs that ‘‘the two-word utterances he [the child] says are neither simple imitations of adult utterances nor random combinations of the words he knows.
- Rather, they follow from the system that the child is using to express meanings at that time.’’
- Several lines of evidence support this view. First, when children first put words together, they tend to combine content words and leave out function words, thus producing utterances such as more milk, push truck, and so on.
- This is similar to the way adults phrase utterances when sending a telegram, where there is a premium on word cost: lost money, send cash, and so on.
- This suggests that the child has an understanding of this grammatical distinction as well as an intuitive appreciation that content words may be more informative than function words.
- Second, as children put words together, particular words are put in particular positions in the sentence.
- A child, for example, is much more likely to say all gone sock than sock all gone. Thus, the child is not merely stringing together separate words that she knows but is putting them together in a systematic way.
- Interpretations of early multiword utterances: What, then, is the child’s system? Several different possibilities have been explored.
- Consider a simple utterance such as baby cry.
- We can describe this in syntactic terms as a subject followed by a predicate. Alternatively, we can describe it in semantic terms as an agent (an animate being who is the instigator of an action) and an action.
- Or we can describe it in positional terms, with baby being a word typically in the initial position and cry as typically in the latter position.
- These characterizations differ in degree of abstractness, with the syntactic description as most abstract and the positional description as least abstract.
- The syntactic description does not appear to fit children’s utterances, at least not in the earliest stages.
- The subject of a sentence may be an agent, but it could also be an object (The book is on the table), an instrument (The nail pierced the wood), or a location (Dallas is dull).
- Acquiring grammatical categories: Ultimately children must grasp categories that are defined in syntactic terms, and there has been much debate concerning how they do this.
- One suggestion is that they use their knowledge of semantic relations to learn syntactic relations.

- This process is known as semantic bootstrapping.
- As Bowerman puts it: children launch their syntactic careers by learning simple order rules for combining words which in their understanding perform semantic functions such as agent, action, and object acted upon, or perhaps other even less abstract semantic functions.
- Through additional linguistic experience, a child may begin to recognize similarities in the way different semantic concepts are formally dealt with and to gradually reorganize his knowledge according to the more abstract grammatical relationships which are functional in the particular language he is learning.
- For instance, children ordinarily use sentences in which the grammatical subject is the semantic agent.
- Then they use this correspondence to begin learning the grammatical category of subject.
- As children become more linguistically experienced, they induce grammatical concepts from the semantic-positional configurations already acquired.
- Exactly how this is done is still very much up in the air, but Maratsos has provided evidence that children acquire some of the concepts during the preschool years.
- Maratsos suggests that children do this by paying attention to the grammatical operations that given linguistic forms take. For example, although like and fond are similar semantically, like takes the grammatical morpheme -ed, whereas the past tense of fond is formed with the auxiliary be (was fond).

Lesson-20

- Children make impressive strides in their acquisition of grammar in their first 2 to 3 years.
- They develop the ability to form simple, functional utterances such as Daddy chair that express their meaning relatively directly.
- Later grammatical acquisitions are built on earlier accomplishments. In this section, we look at two such acquisitions: grammatical morphemes and more complex sentence constructions.
- Children acquire grammatical morphemes gradually throughout the preschool years. As children acquire morphemes, they use them in productive ways, sometimes producing errors such as over-regularizations.
- Complex syntactic constructions such as negatives, questions, and relative clauses are also developed during the preschool years.
- Ease of acquisition appears to be related to the formal and conceptual complexity of the construction, along with certain processing limitations in the child.While a good deal of our linguistic knowledge is tacit, explicit awareness of linguistic units and processes is essential for writing, reading, and other aspects of language.
- The emergence of linguistic awareness takes place after the child's basic grammatical system is organized, in the late preschool and early school .
- Children as young as 2 or 3 years old are able to tell stories and participate in conversations, albeit in limited ways.
- During the subsequent preschool years, they become more flexible and skilled conversationalists and storytellers.

- They use a greater variety of cohesive devices, learn new genres, adapt their speech to different listeners, and formulate and justify requests of others. As children enter school, they have an impressive repertoire of communication skills.
- The language skills that children bring to the school setting are important because language is the predominant means of instruction in a wide variety of subject matters.
- But the language of the school is different from the language of home and of the playground, and children must adapt to these differences as they enter formal schooling.
- The main focus in school is on oral communication in the classroom, and then discusses the relationships between reading and language development.There is, of course, another major difference between language in the school and language before school.
- Schooled language is increasingly written language, and the demands of written language pose a considerable challenge for most children entering formal schooling. The beginning reader is already a fluent language user.
- Many of the comprehension skills that have been acquired to deal with oral language are also applicable to reading.
- These include the ability to extract the meaning of a sentence,interpret that sentence in a given communicative context, draw inferences from individual statements, and monitor one's own comprehension.
- These may be referred to as general comprehension skills. In addition, learning to read involves mastering other skills specific to the written language.
- These include using eye movements to scan sentences in a text, extracting the visual features of letters and words, reading from left to right on a page (in most languages), and relating printed language to spoken language in some way.
- It is likely that some of these skills may be acquired rather easily, but others may take substantial time and effort.
- What this suggests is that reading involves a variety of skills that are well coordinated only in the mature reader.
- That is, the early reader is consumed with the task of identifying even familiar words in a new and unfamiliar mode.
- Early readers thus are less able to attend to the overall meaning of a text and to apply those comprehension strategies acquired in the acquisition of oral language.
- As children master reading-specific skills, they are increasingly able to bring their substantial repertoire of linguistic skills to bear on the task of reading.

Lesson-21

- The meaning and definition of bilingualism varies tremendously from situation to situation.

- **It differs due to:**

- people living in bilingual regions,
- home language is not the same as their school or business language,
- colonization has imposed another language.
- grew up in homes with two languages.

- When children acquire two languages at the same time, their bilingualism is referred to as simultaneous bilingualism.
- Sequential bilingualism occurs when an individual (child or adult) acquires a second language after already acquiring a native language, also referred to as second-language acquisition.
- Most commonly, children learn two languages simultaneously when they are born into a community that is bilingual. In some communities, bilingualism is simply expected.
- Bilingualism is the norm in many parts of the world and younger children are often regarded as superior language learners.
- Some parents and educators fear that bilingual language exposure may slow children's language development and even cause them to mix or confuse their languages.
- Parents may have difficulty being fully objective when recording their **children's language development.**
- Outstanding utterances may be preserved better than errors.
- Case studies have provided detailed data that inspired subsequent studies of bilingual language acquisition.
- Many children learn a second language after attaining considerable proficiency in their native language.
- Gass and Selinker (2001) point out, the boundaries of child second-language acquisition are somewhat **arbitrary.**
- It is commonly thought that there is a critical period for L2 acquisition and that acquiring a second language after puberty is much more difficult.
- For our purposes, it is generally agreed that child **second-language acquisition extends from about 5 to 9 years,** or after the primary language is essentially acquired but before any possible effects related to a critical period.
- Language is arbitrary, the principle that there is (in general) no relation between a **word and its referent.**
- It may be that bilingual children are in general more attentive to language than monolingual children.

- Vygotsky (1934/1986) has suggested, a bilingual child would ‘‘see [one’s] language as one particular system among many, to view its phenomena under more general categories, and this leads to an awareness of [one’s] linguistic operations.’’
- Another cognitive consequence of bilingualism may be cognitive control, the ability to selectively attend to some stimuli and ignore others.
- A fascinating recent report by Bialystok, Craik, Klein, and Viswanathan (2004) suggests that bilingualism may help to offset age-related losses in **cognitive or executive control.**
- Bialystok et al. (2004) used a task known as the Simon task (Lu & Proctor, 1995).
- The task is based on stimulus–response compatibility and assesses the extent to which a person can ignore irrelevant spatial information.
- In the Bialystok et al. study, investigators presented colored stimuli to the left or right side of a computer screen.
- Individuals are faster to respond to congruent trials than to incongruent trials.
- Bialystok et al. (2004) found that bilingualism was associated with smaller Simon effects in both middle-aged and older adults. The bilingual advantage was greater for the older adults.
- The authors suggest that the use of two languages encourages development of cognitive control mechanisms, such as when one has to suppress a word in one language in favor of another language.
- It was once commonly accepted by scholars that bilingualism led to cognitive impairment.
- the prominent linguist Otto Jesperson stated that ‘‘the brain effort required to master the two languages instead of one certainly **diminishes the child’s power** of learning other things which might and ought to be learnt’’
- Many early psychologists also concluded that bilingualism had a detrimental effect on **children’s intellectual development** and academic performance.
- Hakuta has pointed out, however, many of these studies had serious methodological flaws.
- Many studies failed to control for group differences in socioeconomic status between monolingual and bilingual samples.
- Problems associated with bilingualism may have instead been due to low socioeconomic status.

- These studies did not always ensure that the bilinguals were truly fluent in both languages.

Lesson-22

- A necessary condition is one that must be present in order for language to occur in a normal way.
- A sufficient condition is one that, if present, ensures that language will develop normally. It is rare for a complex behavior to have a single sufficient condition.
- It may have several necessary conditions, none of which is sufficient by itself to ensure a positive outcome.
- Three classes of variables are **environmental, cognitive, and innate factors.** All three classes of variables are needed for a complete account of language acquisition.
- Feral children are those who have grown up in the wild.
- Lane presented a detailed description and analysis of a boy named Victor, who was found in the woods of France in 1797.
- There is a period early in life in which we are especially prepared to acquire a language is referred to as the critical period hypothesis.
- Many investigators who favor the critical period hypothesis suggest that there are neurological changes in the brain that leave a learner less able to acquire a language
- A landmark study was reported by Johnson and Newport (1989) who examined native speakers of Korean and Chinese who had immigrated to the United States at various ages between 3 and 39 years of age.
- The evidence from second-language acquisition research has not provided unequivocal evidence for the critical period hypothesis.
- The advantage that younger learners display in some studies may be due to biological changes assumed in the critical period hypothesis.
- Language development deals with the ways adults speak to young children.
- Adult-to-child language, which has been called motherese, differs in a number of ways from adult-to-adult language.

- Speech to children learning language is shorter, more concrete, more directive, and more intonationally exaggerated than adult-directed speech.

- It would appear that such properties would assist children in their language development, data on this basic question are relatively scarce, and widely different opinions exist on the matter (DePaulo & Bonvillian, 1978; HoffGinsburg & Shatz, 1982; Marshall, 1980; Snow, 1979).